

Shesadree Priyadarshani

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EDUCATION

University of Florida

M.S. Applied Data Science, GPA: 4.0/4.0

Coursework: Advanced Machine Learning, Statistical Inference, Applied Data Science, AI Ethics, Advanced Mathematics

Aug 2025 – May 2027

Gainesville, FL

National Institute of Technology Rourkela

B.Tech. Electrical Engineering, CGPA: 8.26/10

Aug 2019 – May 2023

Rourkela, India

RESEARCH EXPERIENCE

UF Intelligent Clinical Care Center (IC3), University of Florida

AI Research Assistant – I-HEAL Research Group

Nov 2025 – Present

Gainesville, FL

- Profiled transformer inference on **HiPerGator GPUs** with **PyTorch** and **CUDA**; characterized **KV-cache memory**, prefill/decode latency, throughput, batching behavior, and prefix reuse across clinical prompt lengths and concurrent workloads.
- Built reproducible **zero-shot and few-shot LLM evaluations** spanning accuracy, AUROC, calibration, bootstrap confidence intervals, and error cohorts; improved clinical-prediction accuracy from **74% to 88%** and AUROC by **9%**.
- Integrated **RAG and QLoRA fine-tuning** with structured retrieval, embedding search, prompt iteration, and grounding analysis; increased grounding by **30%**, reduced hallucinations by **35%**, and improved reasoning consistency by **18%**.
- Developing an interpretable **mechanistic-neural ODE** model for ICU outcomes, combining physiological priors, learned residual dynamics, graph-structured clinical variables, and sensitivity-based feature attribution.

Engineering Education Department, University of Florida

Graduate Student Assistant

Jun 2026 – Present

Gainesville, FL

- Created an **agentic AI orchestration system** with Claude, **MCP connectors**, and Clay.com; coordinated structured ingestion, multi-step synthesis, validation checks, and an interactive Voilà dashboard for alumni and faculty intelligence.

PROFESSIONAL EXPERIENCE

Bristol Myers Squibb

Decision Analyst I – Data Science and AI Solutions

Apr 2024 – Jul 2025

Hyderabad, India

- Engineered reusable **Python feature-generation and batch-scoring pipelines** across **5+ global data sources** for forecasting, retention, and sentiment modeling; added validation and KPI monitoring, improving forecast accuracy by **15%** and supporting **\$2M+** incremental ROI.
- Led offline model assessment and **A/B/test-control experimentation**; used cohort analysis and statistical testing to isolate performance drivers and translate results into decisions for product and marketing stakeholders.

ZS Associates

Decision Analytics Associate – Multi-Channel Marketing

Jul 2023 – Apr 2024

Pune, India

- Developed **SQL, R, and Python feature pipelines** and predictive analyses for response and segment performance; standardized model-ready datasets and recurring scoring jobs, reducing refresh time by **30%** and manual effort by **20%**.
- Delivered **5+ Tableau decision dashboards** for campaign diagnostics, cohort behavior, and KPI monitoring used by global client teams.

SELECTED PROJECTS & RECOGNITION

Distributed Speculative Decoding and Prefix-Cache Serving | PyTorch, CUDA, Ray, FastAPI

Jul 2026

- Implementing a distributed LLM-serving benchmark with **Ray-orchestrated draft workers**, target-model verification, prefix-keyed **KV-cache reuse**, CUDA-aware batching, and reproducible load generation across context lengths and concurrency levels.
- Measuring **time-to-first-token, inter-token latency, p50/p95 latency, throughput, acceptance rate, GPU memory**, and output parity to quantify speed-quality tradeoffs against autoregressive decoding.
- Adding ablations for draft length, verification batch size, prefix overlap, and cache policy to isolate the sources of throughput gains and quality regressions.

Stateful Inference and Cache Policy Lab | PyTorch, FastAPI, Redis, Docker

Jul 2026

- Prototyping a stateful inference service that reuses encoder and prefix computations through content-addressed caches, request deduplication, and reusable model states exposed through a containerized FastAPI interface.
- Comparing **LRU, TTL, and hybrid eviction policies** under bursty traffic and distribution shift while tracking hit rate, p50/p95 latency, throughput, stale reuse, and a correctness-risk score based on embedding divergence and output parity.
- Stress-testing cache safety with controlled state corruption, semantic drift, and forced misses; implementing deterministic fallback to full recomputation when reuse exceeds a correctness threshold.

Sensitivity-Aware LLM Quantization and KV-Cache Compression | PyTorch, CUDA, Transformers

Jun 2026

- Implementing post-training mixed-precision quantization that ranks attention heads and layers by activation and perturbation sensitivity, assigning **4/8/16-bit precision** to reduce model weights and long-context KV-cache memory.
- Benchmarking perplexity, downstream accuracy, tokens/second, TTFT, and peak VRAM against FP16 and uniform INT8 baselines; profiling dequantization overhead and long-context quality degradation.
- Generating per-layer sensitivity maps and calibration-coverage diagnostics to explain precision assignments and identify layers that require higher numerical fidelity.

Constrained Multi-Agent Reinforcement Learning for Traffic Control | PyTorch, SUMO, Ray RLib

Jun 2026

- Building decentralized signal-control agents with shared policy/value networks and explicit constraints on queue growth, unsafe phase transitions, and intersection fairness; comparing independent PPO, MAPPO, and value-decomposition baselines.
- Evaluating travel time, queue length, throughput, constraint violations, and robustness under demand shifts using curriculum training and domain-randomized traffic scenarios.
- Tracking policy stability and coordination through seed-controlled experiments, reward-component ablations, and per-intersection diagnostics in RLib/TensorBoard.

BrailleAI: Two-Way Voice for the DeafBlind – Winner, Best Use of Orkes

Jun 2026

- Won the **Best Use of Orkes** track at the **UC Berkeley AI Hackathon 2026** for an accessible system combining speech recognition, facial-emotion analysis, LLM orchestration, and physical Braille actuation.
- Orchestrated parallel transcription and emotion workers in **Orkes Conductor**, used Claude to generate concise emotion-first Braille output, and triggered an **ESP32 motor controller** through an HTTP task; instrumented retries and task traces for debugging.

ECG Arrhythmia Classification | Python, PyTorch, 1D-CNN, LSTM

Apr 2026

- Trained a patient-level, leakage-safe model on **MIT-BIH ECG signals**, combining waveform representations, RR-interval rhythm context, and class-weighted loss across five AAMI heartbeat classes; achieved **0.91 recall** on ventricular arrhythmias.
- Deployed an interactive **Streamlit** inference demo with waveform visualization, class probabilities, and error analysis.

Multilingual LLM Safety Evaluation – Course Team Project

Nov 2025

- Collaborated with a five-member course team to build a **120-prompt multilingual jailbreak benchmark** spanning direct translation, cultural adaptation, code-switching, and native-language attacks in Hindi, Tamil, Marathi, and Odia.
- Evaluated LLaMA, Mistral, and GPT-family models with **NLLB-600M**, native-speaker review, toxicity scoring, and attack-success-rate analysis; documented cross-lingual safety and reliability gaps for multilingual alignment research.

Automated PCB Test Plan Generation | Python, IPC-2581, BOM Analysis, LLM

Sep 2025

- Automated IPC-2581 netlist parsing, BOM reconciliation, connectivity checks, and LLM-assisted specification interpretation to generate traceable PCB validation plans; reduced manual validation time by **60%** on complex hardware prototypes.
- Produced component-to-test traceability tables and exception reports for missing specifications, duplicate references, and mismatched BOM/netlist records to support reviewer verification.

TECHNICAL SKILLS

Languages & Systems: Python, SQL, R, CUDA, Linux, Git, Bash, REST APIs, FastAPI, HiPerGator HPC

ML Systems: PyTorch, Hugging Face Transformers, LLM inference, KV caching, speculative decoding, quantization, GPU profiling, batching, distributed serving, Ray, Redis, Docker, Kubernetes

Generative AI & Safety: RAG, QLoRA, LangChain, LangGraph, MCP, agentic workflows, prompt engineering, automated evaluation, calibration, bootstrap analysis, red teaming, jailbreak evaluation, SHAP

Modeling & Data: scikit-learn, 1D-CNN, LSTM, neural ODEs, reinforcement learning, causal inference, A/B testing, PostgreSQL/pgvector, pandas, NumPy, SciPy, Airflow, Databricks, Snowflake, BigQuery